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Article title

Genome sequence of psychrophile *Flavobacterium xanthum* DSM3661 isolated from soil in Antarctica.

Authors

Tricia A. Van Laar, [1#] Karisma Naidu, [1] Arielle Names, [1] Silvestre Garcia Pimentel, [1] Jose Jesus Sanchez, [1] Emma Yu[1]

Affiliations

1. Department of Biological Sciences, California State University, Stanislaus, Turlock, California, USA

Running title

Whole Genome of *Flavobacterium xanthum* DSM 3661

Corresponding author's email address

Address correspondence to Tricia A. Van Laar, tvanlaar@csustan.edu

Abstract

A strain of *Flavobacterium xanthum* was isolated from soil in Antarctica. Here, we analyze the whole genome, identifying standard genomic features, the predicted pathogenic potential through the use of virulence factors like genes encoding the O-antigen, resistance to antibiotics such as glycopeptides, and the production of secondary metabolites.

Announcement

Anti-freeze proteins (AFPs) have been identified in organisms across phylogenetic domains, ensuring their survival in frigid environments through the AFPs recrystallization-inhibiting activity and relatively high thermal hysteresis (1). One such strain is *Flavobacterium xanthum* (DSM3661), a gram-negative, non-pathogenic, rod-shaped, psychrophilic bacterium that was isolated from soil in Antarctica based on morphotype in addition to marine 2216 liquid medium and R2A medium (Oxoid) at 20°C (2). Projects such as this one by the Joint Genome Institute seek to improve the database of type strain genomic sequences by increasing the representation of poorly described microbes (3). Further understanding of AFPs has the potential to advance biotechnology through a broad range of applications such as cryopreservation. Sequencing may provide valuable resources for future research regarding cold-resistance mechanisms utilized by *F. xanthum*.

Details on organism growth and DNA isolation to be provided by DSMZ.

The genome was sequenced by the DOE Joint Genome Institute using Illumina technology to generate a 300 bp insert standard shotgun library. Sequencing using Illumina HiSeq-2500 resulted in paired-end reads of 2x150 base pairs in length. Artifacts that remained from Illumina were filtered out using BBDuk and PhiX. Moreover, reads that either had more than one N (uncertain nucleotides), were shorter than 51 base pairs, or a quality score less than 8 were also discarded. BBMAP was then utilized along with reference genomes for humans, dogs, and cats to minimize any potential contamination. Sequencing resulted in 8,860,202 raw read sequences spanning 1,329 Mbp. SPAdes (v3.6.2) (4) was used during genome assembly and the parameters were the following: —cov–cutoff auto—phred–offset 33 –t 8 –m 40 —careful –k 25,55,95 —12. The annotation was then performed following the standards of the JGI Microbial Genome Annotation Pipeline (5). Genome results had a CheckM2 completeness of 99.99 and CheckM2 contamination of 0.61 (6).

PathogenFinder2 v.0.6.0 predicted that *F. xanthum* has a 0.0463 probability of being a pathogen, indicating non-pathogenic capacity (7). Despite this low probability, Virulence Finder Database v2.0.5 found several virulence factors potentially contributing to pathogenicity including the cap8D gene encoding a protein involved in polysaccharide capsule synthesis, the wbtF gene involved in O-antigen production, and the cesC gene encoding an ABC transporter (8, 9). Using the Comprehensive Antibiotic Resistance Database v4.0.1 and the Resistance Gene Identifier tool v.6.0.5, there was a strict hit related to the vanT gene that is predicted to confer resistance to glycopeptide antibiotics (10). Five biosynthetic gene clusters were then found using antiSMASH v8.0, including one RRE-containing cluster and one beta-lactone/T3PKS cluster (11). These genes suggest that the organism may be better adapted for survival through protective surface structures and antibiotic resistance. Because *F. xanthum* resides in cold temperatures, having genes that encode polysaccharide capsule formation may increase the longevity of individuals. The production of the O-antigen and the ABC transporter suggest that *F. xanthum* has developed antibiotic resistance in response to already harsh conditions.

Table 1: Summary of genomic features of *Flavobacterium xanthum* strain DSM3661.

Genome Feature	**Value**
Genome or Total Scaffold Sequence Length (bp)	3,755,582
Number of Scaffolds	101
Scaffold N50 (bp)	15
Average Fold Coverage	0.354x
GC Content (%)	34.42
No. of Genes	3,462
No. of Protein-Coding Genes	3,279
No. of rRNA Genes	10

No. of tRNA Genes	44

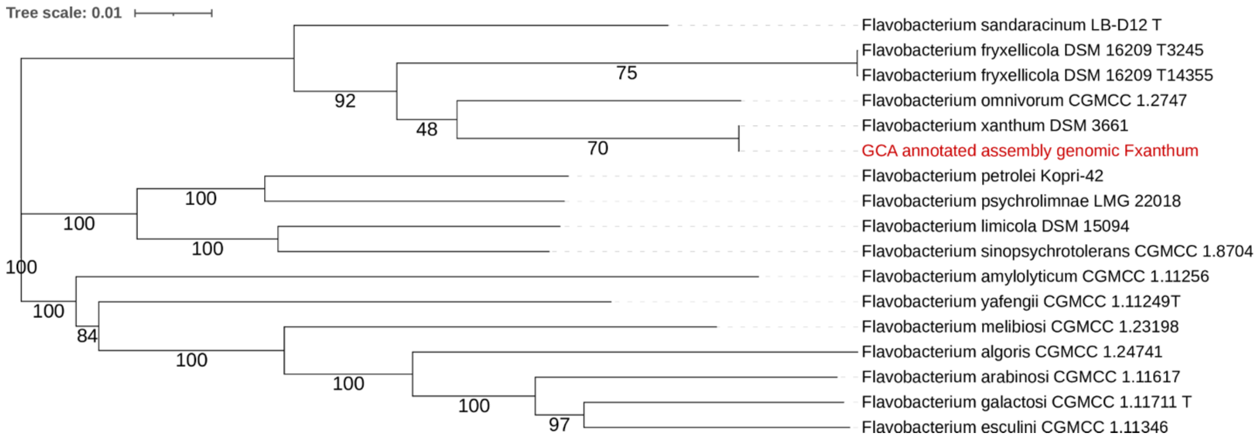

Figure 1: A whole-genome sequence based phylogeny of *Flavobacterium xanthum* (DSM 3661) was generated using the Type (Strain) Genome Server (TYGS). The highest similarity was *Flavobacterium omnivorum* CGMCC 1.2747 (dDDH-d4: 49.8%). The closest relatives were *Flavobacterium fryxellicola* DSM 16209 (dDDH-d4: 43.0%) and *Flavobacterium sandaracinum* LB-D12 T (dDDH-d4: 43.0%) (12, 13).

Data availability statement

This whole genome shotgun sequencing project was deposited in DDBJ/ENA/GenBank using the accession no. FRBU00000000. The version described in this paper is version, FRBU01000000. The raw sequences were deposited in the NCBI SRA under the accession number SRR5188196.

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